Biospectra : Vol. 8(1), March,2013, pp 143-150
An International Biannual Refereed Journal of Life Sciences

Plant Sciences

ISSN : 0973-7057

Ethnobotanical Studies of some plants species used for skin care and skin diseases in Jharkhand

Sunaina Sharma[a]* & A.K. Choudhary[a]
[a]*Tissue culture laboratary, University Department of Botany, Ranchi University, Ranchi

Received 20th January, 2013 Revised 27th February, 2013

Abstract : The paper concern ethno botanical studies on some plants used for skin care or diseases in Jharkhand. Now a day's people are attracted towards the nature and natural medicines due to their less side effects and low cost. Some of the common medicinal plants of Jharkhand which are frequently used by the ethnic people or tribals for treatment of skin diseases by traditional knowledge. Many of these plants are becoming endangered. Some of the plants are listed below.

.**Key words: .**

INTRODUCTION

Jharkhand is undoubtedly a treasure trove of herbal plants and forest produce. A huge

These herbal plants are carrying amazing medicinal / therapeutic value and health only after proper application.

The tribal population and other original inhabitants of Jharkhand have been using herbal plants for the treatment of various skin condition / diseases such as rash, dermatitis, eczema, psoriasis, dandruff, acne, skin abscess (boil or furuncle), herpes, hives, shingle, ringworm, and scabies.

Natural skin care is the way to maintain the skin using naturally derived ingredients (such as herbs, roots, leaves, bark, flower, seeds, oils) and whole plants.

The indiscriminate removal of the bark / whole plant results in the death of the plant / tree, which is the main reason for their depletion from the nature / forest making them endangered species.

*Correspondent author :
Phone :
E-mail :

MATERIALAND METHOD

1. Andrographis paniculata

Common names……Kalmegh, Chiraita, Bhuineeem (N), Bhuinimo (O), Bhuinim (M), Creat, Kariyat (E).

Scientific Classification

Kingdom …… ***Plantae***
Order …… ***Lamiales***
Family …… ***Acanthaceace***
Genus …… ***Andrographis***
Species …… ***A.paniculata***

Uses

- It has been used in the treatment of some skin infections by herbal practitioners of Jharkhand.
- It is considered beneficial to the skin and is used both internally & externally for this purpose.
- It is also used for iches.
- A decoction of the plants is blood purifier.

2. Achyranthes aspera Linn

Common names……Sitirkad, Chitchat (M), Latjira, Chirchiti(H), Apang, Chirchiti (B), Prickly chaff flower(E).

Scientific Classification

Kingdom …… ***Plantae***
Order …… ***Caryophyllales***
Family …… ***Amaranthaceae***
Genus …… ***Achyranthes***
Species …… ***A.aspera***

Uses

- Decoction of the plant is used as laxative and is applied externally on boils and pimples.
- For itching powered root (5g) are taken orally with water twice a day for seven days.
- Poisonous bites……roots (2.5g) are ground into a fine powder. 5gm of this powder is taken orally and remaining paste is applied on the bites**.**

3. Argemone Mexicana

Common names…… Shialkanta, Satyanashi(H), Darudi (G), Brahmadandi(S), Kutiyoti(T)

Scientific Classification

Kingdom …… ***Plantae.***
Order …… ***Papoveraceae.***
Family …… ***Ranunculales.***
Genus …… ***Argemone.***
Species …… ***A.mexicana.***

Uses

- For itching seeds are ground and mixed in coconut oil and applied on the infected body parts.
- Whole plant is crushed along with 2 nos. of black pepper and the paste is applied tropically.
- Juice extracted from roots as administered orally for poisonous bites.
- Used for wounds.

4. Bombax ceiba

Common names…… Shalmali, Semal(H), Silk Cotton tree(E), Tera(M)

Scientific Classification

Kingdom …… ***Plantae.***
Order …… ***Malvales***
Family …… ***Malvaceae***
Genus …… ***Bombax***
Species …… ***B.ceiba***

Uses

- Paste of thorns and milk applied on the pimples for 7 days gives good results.
- For wounds fresh bark is ground, mixed with little



Fig.1: ***Andrographis paniculata***



Fig.2: ***Achyranthes aspera Linn***



Fig.3: ***Argemone Mexicana***

water & applied on infected part.

- It is applied on swelling, boil and burning sensation.
- A paste made out of prikles is good for restoring skin color especially on face.

5. Boerhavia diffusa

Common names ……. Snathikari(H), Punurnava(B),Dholia-Saturdo **(G),** ambadivasu (M).

Scientific Classification

Kingdom …… ***Plantae.***
Order …… ***Caryophyllales.***
Family …… ***Nyctaginaceae.***
Genus …… ***Boerhavia.***
Species …… ***B.diffusa.***

Uses

- The paste is applied on the wounds, dries up the oozing.
- The root of the plant is an effective remedy for several skin diseases.
- A paste of the root can be applied as a dressing for oedematous swellings.
- A hot poultice of the root can be applied with gratifying results to ulcers, abscessis and similar skin diseases
- It is also used for extracting guinea-worms.
- It is also used for leprosy.

6. Butea monosperm

Common names … Palash, Dhak (H), Flame of the forest (E), Parrot tree(E), Keshu(P), Kesudo (G).

Scientific Classification

Kingdom …… ***Plantae.***
Order …… ***Fabales.***
Family …… ***Fabaceae.***
Genus …… ***Butea.***
Species …… ***B.monosperma.***

Uses

- Paste of the powdered seed with lemon juice is applied as a cure for ringworm and herpes.
- Flowers are used to cure leprosy, skin diseases, swelling and pimples.
- Flowers are boiled in water to give fragrant, deep yellow water which has medicinal properties and prevent skin problems in the coming summer.

7. Cassia tora

Common names…….. Chakvad(H), Chakunda(B), (O), Kawaria **(G)**



Fig.4: ***Bombax ceiba***



Fig.5: ***Butea monosperma***



Fig.6: ***Cassia tora***

Scientific Classification
Kingdom …… ***Plantae.***
Order …… ***Fabales***
Family …… ***Fabaceae.***
Genus …… ***Senna***
Species …… ***C.tora***
Uses

- The juice extracted from its leaves is used in case of skin ailments, rashes and allergies.
- The leaves and seeds are useful in leprosy.

8. Hydrocotyle asiatica

Common names…… Brahma manduki (H), Indian pennywort (E), Gotu Kola
Scientific Classification
Kingdom …… ***Plantae.***
Order …… ***Fabales***
Family …… ***Fabaceae.***
Genus …… ***Senna***
Species …… ***C.tora***
Uses

- Keratinization (the process of replacing skin after sores or ulcers).
- Promotes healthy skin.
- Stimulates the formation of lipids and proteins necessary for healthy skin.
- Used for second and third degree burns.
- It cures wounds.

9. Solanam torvum

Common names……Bhurat,Bhankatiya(H), West Indian Turkey Berry (E), Brihati(S), Ran-Baigan, Goth-Began (Folk).
Scientific Classification
Kingdom …… ***Plantae.***
Order …… ***Solanales.***
Family …… ***Splanaceae.***
Genus …… ***Solanum.***
Species …… ***S.torvum.***
Uses

- Leaves are used for treating skin diseases.
- Roots in cracks in feet.
- It is used for the treatment of pimples and wounds.
- It is useful in leprosy.



Fig.7: ***Hydrocotyle asiatica***



Fig.8: ***Solonam torvum***

10. Psoralea corylifolia

Common names …… Babchi (H), Bakuchi(S), Lata kasturi, Somaraji, Vakuchi.
Scientific Classification
Kingdom …… ***Plantae.***
Order …… ***Fabales.***
Family …… ***Fabaceae.***
Genus …… ***Psoralea.***
Species …… ***P.corylifolia.***
Uses

- Seed oil is extremely beneficial in numerous skin ailments.
- In hypopigmented lesions in the skin like leucoderma and psoriasis respond well to local application along with oral therapy.

- In scabies and ringworm infestation the bukuci seed powder mixed with buttermilk is applied externally .
- In chronic skin diseases a mixture of bakuci and karanj oil is commonly used with Vaseline, scabies , psoriasis, ringworm are treated successfully with bakuci.

11. *Madhuca longifolia*

Common names**……..**Honey tree, Butter tree (E), Mahua (H), Maul (B), Mohwa (M), Ippa (T).

Scientific Classification

Kingdom …… ***Plantae.***
Order …… ***Ericales..***
Family …… ***Sapotaceae.***
Genus …… ***Madhuca.***
Species …… ***M.longifolia.***

Uses

- Decoction of stem bark is used to cure skin disease.
- Powdered bark is employed for the treatment of scabies.
- Bark heals wounds.
- The leaves are applied as a poultice to relieve eczema.
- Bark is used in chronic leprosy.

12. Millettia pinnate

Common names……Indian beech (E), Karanj,Karanja(H),(B)(M),(G),Ponga, Pongam(T), Naktamala(S).

Scientific Classification

Kingdom …… ***Plantae.***
Order …… ***Fabales..***
Family …… ***Fabaceae.***
Genus …… ***Millettia.***
Species …… ***M.pinnate.***

Uses

- It is used for skin protection, as it is believed to enhance the **uv** absorbing properties of conventional sunscreams.
- Karanj infused baths are used for cleaning skin ulcer and sores.
- Paste is used for treating skin diseases including acne, herpes, and leucoderma (partial or total loss of skin pigmentation, often occurring in patches.)
- Bark is useful in skin alinments and scabies.
- Bark also yields a black gum that is used to treat wounds caused by poisonous fish.



Fig.9: ***Madhuca longifolia***



Fig.10: ***Millettia pinnate***

13. ***Jatropha curcus***

Common names……Physic nut , purging nut(E), Ratanjot , Jangali erandi(H), Jepal (G),Katamanak (M),Kananaranda(S).

Scientific Classification

Kingdom …… ***Plantae.***
Order …… ***Malpighiales.***
Family …… ***Euphorbiaceae.***
Genus …… ***Jatropha.***
Species …… ***J.curcas.***

Uses

- The oil from seeds is used to cure rashes and parasitic diseases.
- Sap from the bark is used to clear wounds and ulcers and also used to stop bleeding .
- Seed oil is effective against scabies.
- Sap from the leaves used on bee and wasp stings.
- Paste made from leaves is applied externally for smooth skin .
- Latex is applied on the burnt part.

14. Vernonia cinerea

Common names…….. Sahadevi, Sadodi (H), Blue Fleabane, Strongman-Bush(E), Sadori(G), Mukuthipundu(Tam),Gherittekarnina(Teg)

Scientific Classification

Kingdom …… ***Plantae.***
Order …… ***Asterales.***
Family …….. **Asteraceae.**
Genus …… ***Vernonia.***
Species …… ***V.cinerea.***

Uses

- Decoction of the plant is used to treat various skin diseases.
- Decoction is used to dress wounds.
- This plants can be used to cure boils, psoriasis.
- Seeds(5g) are eaten with warm water to cure psoriasis.

15. ***Ziziphus Mauritian***

Common names……Indian jujube, Indian plum (E), Ber(H), Badri (S), Elandhai (Tam).

Scientific Classification

Kingdom …… ***Plantae.***
Order …… ***Rosaless.***
Family …….. **Rhamnaceae.**
Genus …… ***Ziziphus.***
Species …… ***Z.mauritiana.***

Uses

- Fresh leaves (100g) are boiled in one liter of water. The mixture is cooled and used to wash hair.
- Leaves are beneficial for hair loss, grey hair and helps to strengthen the hair.
- Paste is prepared from the leaves and applied tropically to cure acne.
- Bark paste is applied on sores.
- Powdered root is dusted on wounds.

16. ***Vitex negundo***

Common names : Common chaste (E), Sindwar, Nirgundi(H), Sindhuvara (S), Nishinda(B).

Scientific Classification

Kingdom …… ***Plantae.***
Order …… ***Lamiales..***
Family ……... ***Lamiaceae.***
Genus …… ***Vitex.***
Species …… ***V.negundo.***

Uses

- Seeds make cooling medicine for skin diseases and leprosy.
- Mild skin rashes with itching can be cured.
- Root is used for worms, boils and skin disorders.



Fig. 11: ***Vitex negundo***

OBSERVATIONS AND DISCUSSION

This is evident from the present study that there exists a wealth of traditional knowledge about medicinal uses of local flora in Jharkhand The tribal population and the other original inhabitants of Jharkhand have been using these medicinal plants for the treatment of various diseases such as skin disease and other diseases . Traditional medicine has an established promotive, preventive, curative and rehabilitative role. It can be the main form of health care or an integrated component of mainstream health care or an alternative or complement to the main form of health care (WHO 2001).

The qualitative analysis of phytochemicals encompasses steroids, alkaloids, flavonoids triterpenoids,

tannis, quinine, coumarin, protein sugar .The presence of these various phytochemicals shows the importance of A.paniculata and B. diffusa which could be used to cure various skin ailments.

There has been a report on comparative protein profile of granulation tissues of burn, diabetic and immunocompromised wounds treated with 5.0% (w/w) ointment of methanol extract of the plant.

Methanolic leaf extract of A.aspera as constituents of a poly herbal hair oil (PHO) showed anti-dandruff activity.

Ghosh and his group (2005) studied in vivo wound healing activity of the extract and the latex of A.mexicana on excision wound healing model- the result demonstrated significant wound healing activity of the test extracts that is comparable with the established drug nitrofurazone, the tensile strength of the extract treated group was found to be higher than the latex treated group of animals on 12th post wounding day (Ghosh et al, 2005).

Significant wound healing activity of petroleum ether and butanol fractions of ethanol extract of A.mexicana containing some sterols, alkaloids proteins and carbohydrates was also reported in albino rat model by Patil and his group (2001). Dash and Murthy (2011) investigated wound healing activity using excision, incision and dead space wound models in wistar albino rats with different extract of *A.mexicana* leaves.

The isolated steroids from the plant also have been used to treat leprosy upon treatment with centella asiatica , maturation of the scar is stimulated by the production of type I collagen .Centella asitica is a profusely branched prostrate herb consisting of active principles such as *Vallrine ,Asiaticoside ,Sitosterol, Tannin , Oxy-asiaticoside. Asiaticoside* is used in the treatment of leprosy.

Patkar [34] refers to the formations described in *Anubhoga Vaidya Bhaga, a* compendium of formulations in cosmetology, in outlining the use of vitex leaves along with those of Azadirachatanindica, Eclipta Alba, Sphaeranthus indicus and carum copticum in a notable rejuvenation treatment known as Kayakalpa.

Methanolic extract of vitex negundo may be useful to be applied as an effective biomedicine for the treatment of scarbies in people skin.

The medicinal plants used by tribals for the treatment of diverse ailments by traditional knowledge. Many of these plants are becoming endangered .Scientific studies should be conducted on these plants .It is extremely important to save this traditional knowledge of biological heritage and explore new resources.

REFERENCES

1. **Ambasta, S.P., 1986.** The useful plants of India, Publication and information: *CSRI New Delhi*.
2. **Bhamre, P.B., 1998.** Traditional Knowledge of Plants for skin ailments of Dhule & Nandurbar districts Maharaashtra (India), *J. Phytol R*es. **11**(2): 195
3. **Bhandri, M.J., Chandrashekar, K.R. and Kaveriappa, K.M., 1995.** Medicinal Ethnobotany of Siddis of Uttar Kannada district Karnataka, *J, Ethnopharmacol,* **4**(7): 149.
4. **Bhattari, N.K., 1993**. Folk Medicinal used of plants for respiratory complaints in central Nepal, *Fitoterapia*, **66** (2):163-170.
5. **Chandra, K., 1985.** Traditional remedies of Bahraiah and Gonda districts of Uttar Pradesh, *Sachitya Ayurveda*, **37**: 483 - 486.
6. **Chantia, A., 2003**. Traditional Knowledge of ethno medicine in Jaunsarbawar, Dehradun district, *Indian J Trditional Knowledge*, **2**(4): 397-399.
7. **Chopra, R.N., Nayar, S.L. and Chopra, I.C., 1956.** Glossary of India Medicinal Plants. (Publications and informations Dirctorate New DeIhi),
8. **Jain, A., Katewa, S.S., Galav, P.K. and Sharma, P., 2005.** Medicinal plant diversity from the sitamata wild life sanctuary, chittoragarh district India, *J. Ethnopharmacol,* **102**(3):543-557.
9. **Ghosh, *et al* (2005), Dash and Murthy(2011).**Investigated wound healing activity in A.mexicana.
10. **Medicinal Plants and Raw Drug of India –** By Purshotam Kaushik & Anil
11. **Indian medicinal Plants-**By orient Longmann.
12. **Medicinal Plants-**Robert Bentley & Henry Trimen.
13. **Medicinal and Aromatic Plants-**E.John Jothi Prakash.

* * *